

1. Low back pain: initial assessment (>16y)

Based on BMJ 2019;367:l6273, NICE 2020 NG59, JAMA 1992; 268:760

Assessing low back pain: what are we trying to achieve?

- Decide which of the 3Ns it is:
 - 'Nasty' back pain (red flags).
 - Nerve compression back pain (sciatica-like pain or spinal stenosis).
 - Non-specific low back pain.
- Refer and investigate **if appropriate**. NICE reminds us that we should not be imaging in primary care.
- Identify those at high risk of chronic pain using risk stratification, e.g. STarT Back tool, and adopt a biopsychosocial approach to tackle this.
- Share decisions about appropriate management options, with particular emphasis on active management, e.g. self-care and keeping active.
- Safety-net. *Specific information for each of the 'Ns' is covered in the subsequent pages.*

INITIAL ASSESSMENT OF BACK PAIN – the 3Ns

	'Nasty' back pain (red flag and likely to need urgent referral)	Nerve root compression (sciatica or radicular pain)	Non-specific low back pain (85–90% of what we see!)
Features in the history	Red flags present, including: Trauma. Unexplained weight loss. Neurological symptoms (progressive or new bowel/bladder incontinence). Age <20y or >55y. Fever. IVDU/Inflammatory pain. Steroid use. History of cancer (prostate, kidney, thyroid, breast).	No red flags. Leg pain more severe than back pain (usually unilateral). Neuropathic-sounding pain. Numbness/paraesthesia of lower leg. Positive neural tension test (see below). Neurological deficit corresponding with nerve root affected.	Clinical diagnosis based on: - No red flags. - No features of radicular pain.
Examination	Do a neurology examination. Consider assessing sphincter tone if neuro red flags. There are no sensitive/specific exam findings for cancer: refer based on history if suspected.	Do a neurology examination. Do a straight leg raise on both sides (<i>pain on the affected side when <u>opposite</u> side lifted very specific for disc prolapse</i>). Consider a slump test if straight leg raise unclear.	Neurological examination not necessarily needed (no features of radicular pain/red flags means <1/1000 chance significant disc prolapse). Consider examining hip joints and for muscle tenderness.
Possible causes	- Fracture (trauma/osteoporosis). - Malignancy. - Infection, e.g. discitis or osteomyelitis. - Cauda equina/spinal stenosis. - Inflammatory back pain.	- Sciatica (usually disc prolapse between L4 and S1). - Much rarer compressive lesions, e.g. tumour, abscess.	- May be sprain or strain. - Often unable to identify an anatomical driver (or nociceptive focus).
Action	Refer to secondary care. Urgency will depend on symptoms and suspected condition, but may include: - A&E. - MSCC coordinator. - Suspected cancer pathway. - Urgent MSK referral.	Risk assess, analgesia, safety-net. Risk stratify with STarT Back. Refer urgently if red flags or progressive neurological deficit. Most managed conservatively. Consider referral for intervention according to your local pathways.	Advice, education, reassurance. Risk stratify with STarT Back and explore biopsychosocial context. Give a clear explanation. Promote active treatment/self-care. Safety-net/give cauda-equina card. Do not image.

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. June 2025. For full references see the relevant Red Whale articles.

2. Acute 'nasty' (red flag) back pain: identification and management

'Nasty' cause	When to suspect and what to do!	
Acute spinal cord compression or pain suggestive of spinal metastases <i>Contact MSCC (metastatic spinal cord compression) coordinator</i>	The spinal cord itself is compressed, usually by trauma/cancer/abscess or haematoma. Can occur up to any level, but remember that spinal cord ends at L1/2 (NEJM 2017; 376:1358). Consider if: • Past or current history of cancer, with progressive back pain, tenderness or bilateral sciatica. • Gait disturbance or 'heavy legs'. • Lower motor neurone pattern AT THE LEVEL of the lesion = EVERYTHING goes down! (weakness, reduced tone, down-going plantars, reduced/absent reflexes) • Upper motor neurone signs BELOW the level of the lesion = EVERYTHING goes up! (increased tone, up-going plantars, brisk reflexes) • New-onset urinary incontinence or retention. • May have a sensory level. • See our article on <i>Spinal cord compression and cauda equina syndrome</i> for more details.	
Cauda equina syndrome (CES) <i>New or progressive CES symptoms ≤14d = A&E</i> <i>Stable CES symptoms >14d OR bilateral sciatica = urgent MSK clinic <2w</i>	See our article: <i>Spinal cord compression and cauda equina syndrome</i> Central disc prolapse is the most common cause, but can be any of trauma, cancer, abscess or haematoma. We are aiming to pick these up EARLY, before urinary incontinence occurs. Bilateral sciatica occurs in only around 50% of cases. Anal tone and saddle sensation are not reliable enough to use as exclusion tests if normal (BMJ 2021;372:1). Use cauda equina cards (link below) and look for:	
	Early signs (more subtle) • Reduced sensation of micturition or altered saddle sensation (pins and needles/tingling). • Difficulty/delay in opening bladder/bowels. • Change in erections/vaginal sensation (this may be the only symptom – ask!).	Late signs • Saddle anaesthesia or reduced anal tone. • Incontinence (this is overflow – people go into retention first). • Weakness. • Lower motor neurone signs.
Infection <i>Refer immediately</i>	For example, discitis/osteomyelitis/abscess. Relatively uncommon. Consider in IVDU or long-term steroids/immunosuppression, PMH of TB, recent surgery or recent systemic infection/sepsis. Consider if: • Fever/malaise/systemically unwell. • Increasing and unremitting back pain. May be aggravated by straining/have focal bony tenderness. • Neurological symptoms/signs (only present in 10–15%).	
Trauma <i>Refer immediately via ambulance (for imaging) if any of these features are present</i>	If you are asked to assess a patient with a history of traumatic back pain, we can use the NICE 2016 NG41 guidance to identify who needs urgent imaging and immobilisation; most of us will need to call the paramedics to achieve this. NICE identifies high-risk features as: • Age 65y or older and reporting thoracic/lumbosacral pain. • High-risk injury, e.g. fall from height >3m, high-speed or rollover motor collision, ejection from a motor vehicle, axial load injury (e.g. falling onto feet/buttocks), bicycle or horse-riding accidents. • Known or high risk of osteoporosis. • Abnormal examination findings, e.g. neurological signs, new deformity or midline tenderness, spinal pain on coughing or sneezing (impulse pain), focal bony tenderness. • Pain or abnormal neurology when mobilising (stop mobilising immediately if this occurs!). To assess C-spine acutely, NICE suggests using the Canadian C-Spine rules (link in useful resources).	
Inflammatory back pain (axial spondyloarthritis)	Suspect inflammatory back pain in people with low back pain lasting more than 3m, starting before the age of 45y AND with <u>4 or more</u> of these criteria (NICE 2017; NG65): • Onset before age 35y. • Waking in the second half of the night with symptoms. • Buttock pain. • Improvement with movement. • Improvement within 48h of taking NSAIDs. • First-degree relative with spondyloarthritis. • Current or past arthritis or enthesitis (pain/inflammation at insertion of tendons, e.g. plantar fasciitis). • Current or past psoriasis. Referral: • If 4 or more criteria present, refer to rheumatology (no further testing required). • If 3 criteria are present, do an HLA-B27 test and refer to rheumatology if positive. • If fewer than 3 of these criteria are met, advise patients to re-present if they have new symptoms, particularly if they have a past or current history of inflammatory bowel disease, psoriasis or uveitis.	

3. Sciatica: assessment and management

BMJ 2019;367:l6273, NEJM 2020; 382:12:1093, NICE 2020 NG59

Red Whale



GEMS

Guidelines & Evidence Made Simple

Typical presentation

Typical history features are:

- Unilateral leg pain (aching or sharp) **WORSE** than back pain.
- Pain radiates to back of leg/below knee (may be anterior thigh if L4).
- Numbness/paraesthesia in the lower leg.

Typical examination findings:

- Positive straight leg raise/slump test.
- May have neurological deficit in the affected myotome/dermatome.

Differential diagnosis/risk modification

Nerve root compression

- **Disc herniation at L4/5 or L5/S1 by far most common.**
- Consider lumbar spinal stenosis, particularly if bilateral and age 60y+. Seek history of neurogenic claudication (*see separate article on Lumbar spinal stenosis*).
- Soft tissue stenosis, e.g. tumour, infection, cysts.

Other conditions can mimic the symptoms of sciatica:

- Sacroiliac joint pain.

Modifiable risk factors: smoking, obesity and manual work.

MRI imaging and referral

NICE states that we should not image in primary care. Secondary care will usually do this if referral is needed.

Why? Disc herniation on MRI is a common age-related finding and does not correlate well with symptoms (e.g. in a recent study, 57% of symptomatic people compared with 34% of asymptomatic people).

Consider referral if:

- Symptoms fail to improve, with ongoing pain and impact on function (*see your local pathways for timescales*).
- Severe or progressive neurological deficit is present, e.g. weakness, wasting, foot drop.
- Suspicion of cauda equina syndrome, or bilateral sciatica.
- Other red flags: history of trauma, suspicion of cancer, previous cancer or suspected infection.

Management

Risk stratify with STarT Back: to prioritise resources (*see next page*).

Safety-net: offer cauda equina card (*see resources*).

Self-care: keep active and avoid bed rest.

Physiotherapy or manual therapy.

Analgesia: NSAIDs (limited evidence) or mild opiate (only in first 12w, not beyond). Neuropathic agents: NICE low back pain/sciatica guidance says do NOT use gabapentinoids or benzos at all, and do not use opiates in chronic (>12w) sciatica. Discussed more in main article.

Epidural injection of local anaesthetic/steroid. NICE offers as an option; limited evidence of short-term benefit, but may be enough to restore function while nerve irritation resolves (*see drug dilemmas section*).

Surgery: usually microdiscectomy (endoscopic or open).

Most people

Few people

Prognosis

Studies of prognosis report very different results:

- A secondary care study reported that 90% of patients receiving conservative care will improve within 4 months of symptom onset.
- In contrast, a UK primary care cohort study of 452 patients suggests that only 55% report improvement in pain/disability by 1 year. 11% were referred to secondary care for injection or surgery.

Benefits of surgery?

NICE: consider spinal decompression if symptoms persist and are consistent with radiological findings. Optimal timing is not clear; imperfect studies suggest:

- Surgery within 6–12w of onset of radicular pain provides faster pain relief than conservative management.
- By 12–24m, there is no difference in outcomes between surgical/conservative management.
- An RCT of people with >4m of radicular pain suggests less pain at 6m in those having microdiscectomy vs. conservative management (NEJM 2020;382:12:1093).

4. Non-specific low back pain: initial risk stratification and management

NICE 2020 NG59, DTB 2019;57(7):104

Risk stratify using the STarT Back tool

NICE recommends that we use the STarT Back tool because it gives us a clear framework to identify which treatments to use for people presenting with non-specific low back pain. It helps us identify those at greater risk of developing persistent pain/disability and support them earlier.

Best administered electronically ([link in useful resources box of main article](#)).

Why do it? Directs resources such as physiotherapy and psychological input more appropriately than clinician usual care.

STarT Back tool questions and scoring (if you don't have access to electronic form!)

Ask the following and score one point for each positive answer (except Q9):

Q1: Has your back pain spread down your leg(s) at some time in the last 2 weeks?

Q2: Have you had pain in your shoulder or neck at some time in the last 2 weeks?

Q3: Have you only walked short distances because of your back pain?

Q4: In the last 2 weeks, have you dressed more slowly than usual because of your back pain?

Q5: Do you think it's not really safe for a person with a condition like yours to be physically active?

Q6: Have worrying thoughts been going through your mind a lot of the time?

Q7: Do you feel that your back pain is terrible and never going to get any better?

Q8: In general, have you stopped enjoying things you usually enjoy?

Q9: Overall, how bothersome has your back pain been in the last 2 weeks?

(Not at all (0), slightly (0), moderately (0), very much (1), extremely (1))

You then calculate a **TOTAL SCORE** and a **SUB-SCORE** (based on Q5–9).

Low risk of chronicity = total score of 3 or less.

Medium risk of chronicity = total score of 4 or more AND sub-score of 3 or less.

High risk of chronicity = total score of 4 or more AND sub-score of 4 or more.

Safety-net and document all for cauda equina syndrome (see link for cards in 28 languages)

Low
risk

Treat in primary care. Offer:

- Reassurance.
- Information, e.g. VA back pain leaflet ([link in resources](#)).
- Guided self-help.

Medium
risk

Refer to physiotherapy. Offer:

- Reassurance, information, guided self-help AND
- Physiotherapy (6–8 sessions)/ manual therapy.

High
risk

Refer to physiotherapy/pain clinic for:

- Combined supported exercise/ physiotherapy.
 - Psychological approach.
- Many MSK pathways now have 'back school' or equivalent.*

Drug treatments

For non-specific low back pain, NICE recommends:

- NSAIDs +/- gastroprotection if no contraindication at lowest dose, shortest duration possible.
- Do not offer paracetamol ALONE (no more effective than placebo).
- Consider weak opiates if NSAIDs contraindicated/ineffective (*acute script only with clear discussion of goals of treatment and plan to stop*).
- Do NOT use anticonvulsants or antidepressants.

NICE states that there is no role for spinal injection, but that facet radiofrequency denervation may be considered if moderate to severe pain and non-surgical treatment has failed.

Investigations and treatments to AVOID

NICE recommends we do NOT offer or recommend (on the basis of absence of evidence):

- Traction.
- Bed rest.
- Orthotics/belts/corsets.
- Ultrasound or TENS.
- Acupuncture (*though the DTB disagrees*).